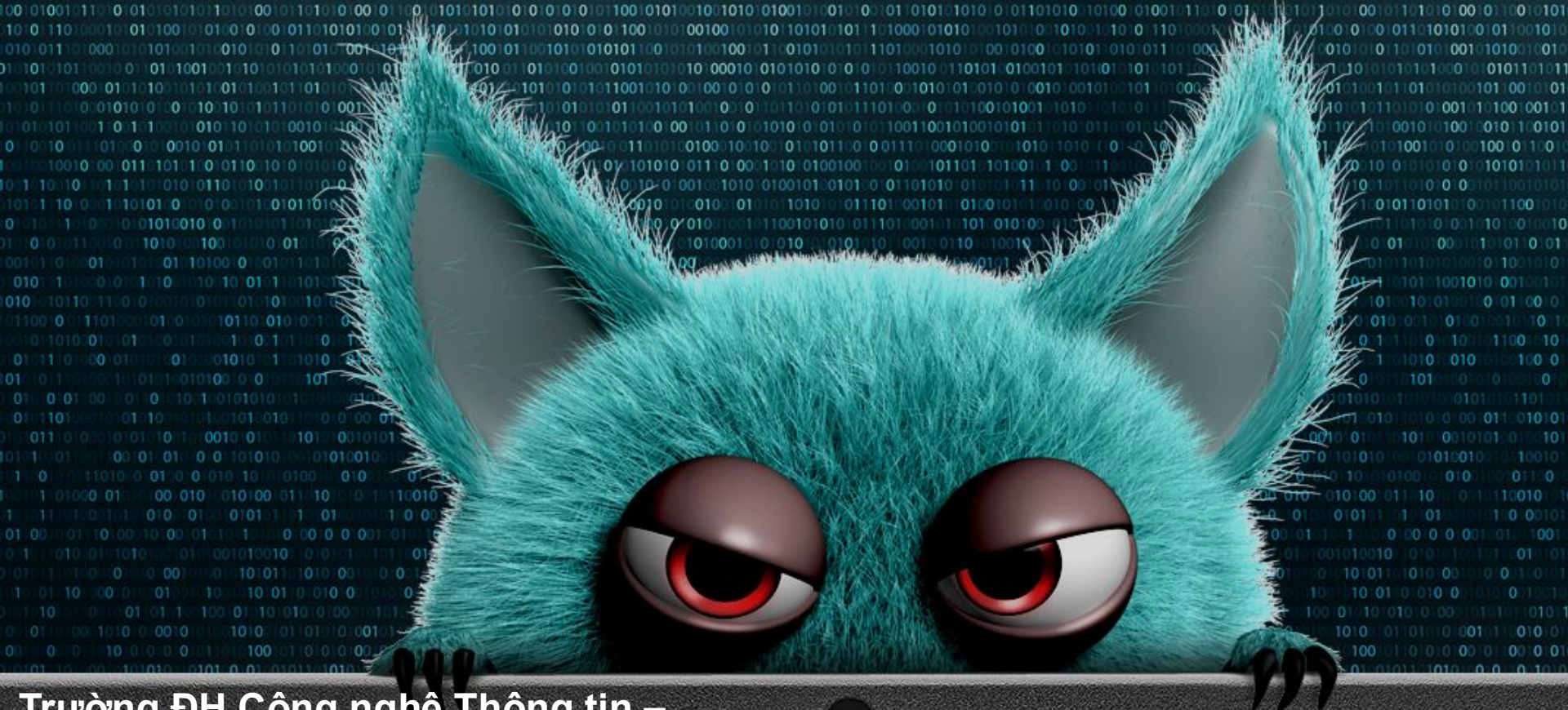




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ĐHQG TP. HCM



Cơ chế hoạt động của mã độc

NT230 – Malware's Modus Operandi



NT230 - Cơ chế hoạt động của mã độc



Bài giảng lý thuyết – 09.2025
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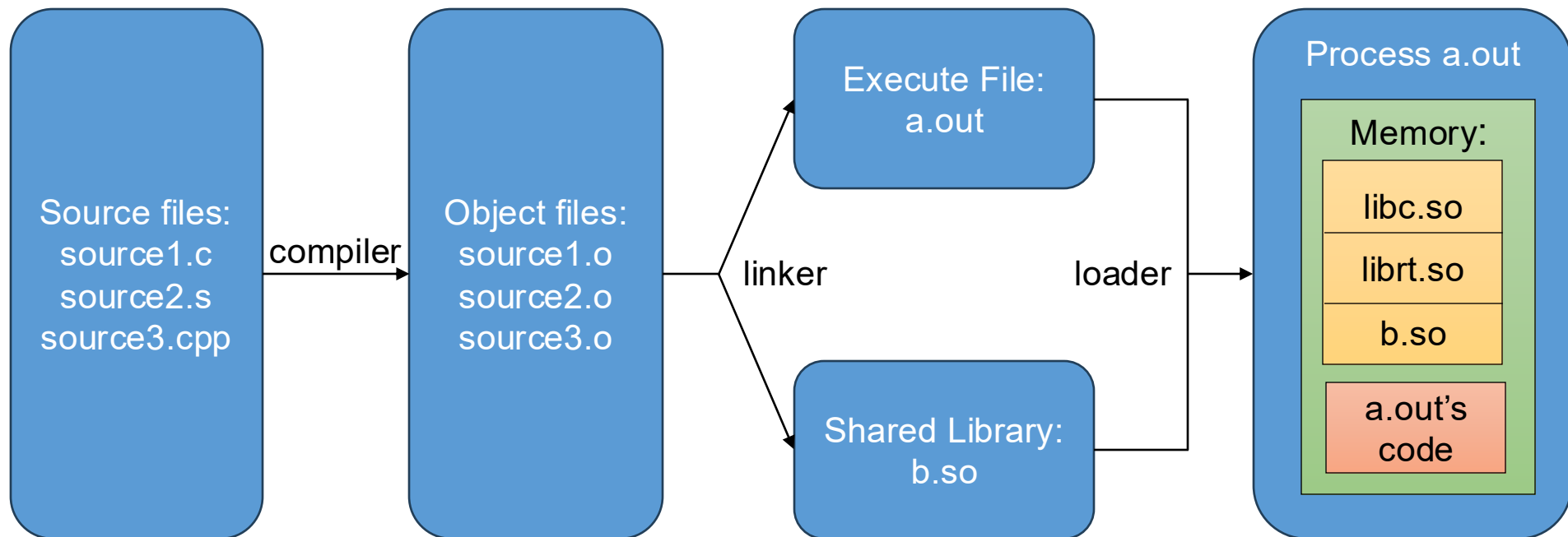


EXECUTABLE FILE FORMAT

WEI WANG – Defense Against the Dark Arts



Compiler, Linker and Loader



- Compiler transforms source code into binary machine code (object code)
 - Example: gcc, Clang, vc_compilerCTP.exe
- Linker takes object files and libraries files, and combines them into a single executable file or library file
 - Example: GNU ld, lld, LINK.exe
- Loader load an executable file and libraries into memory to start a new process (part of OS)
 - Executable loader: load executable files
 - Example: `execve` (system call)
 - Dynamic linking load: load dynamic libraries
 - Example: `ld-linux.so`

Compiler, Linker and Loader cont'd

More about Loader



- Brings an executable file and required libraries on disk into memory to start a new process
- Tasks:
 - **Copy** executable file code (text section) and global variables (data section) into memory
 - Copy arguments and environment variables into memory
 - Initialize registers
 - Jump to start of program to execute (`_start` function)
 - Load dynamic libraries (**map** dynamic libraries code into memory)



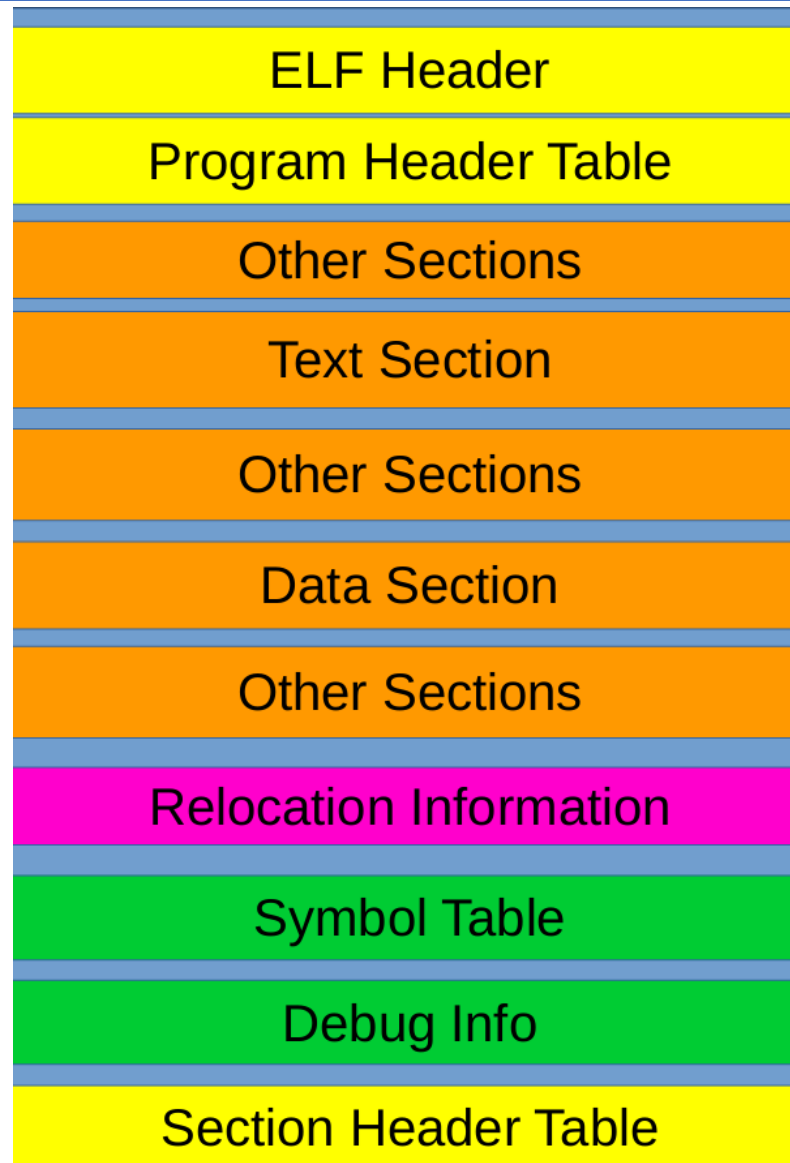
- For compiler, linker and loaders to work properly, they have to agree on the format of object files, executable files and library files
- The most common formats are:
 - ELF on *nix: Executable and Linkable Format
 - PE on Windows: Portable Executable
 - Mach-O on OS X

- Executable and Linkable Format
- Defines format for:
 - Executables
 - Object files
 - Dynamic libraries (shared libraries)
 - Core dumps

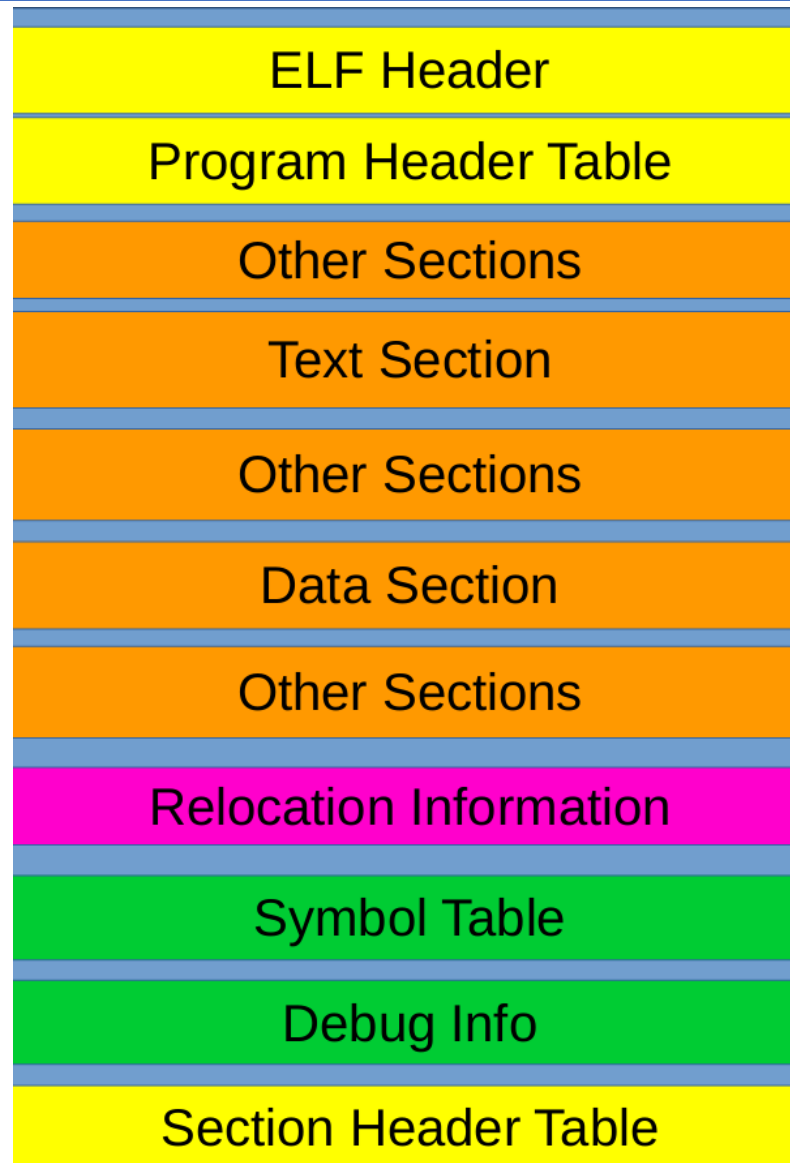
ELF Format Examples



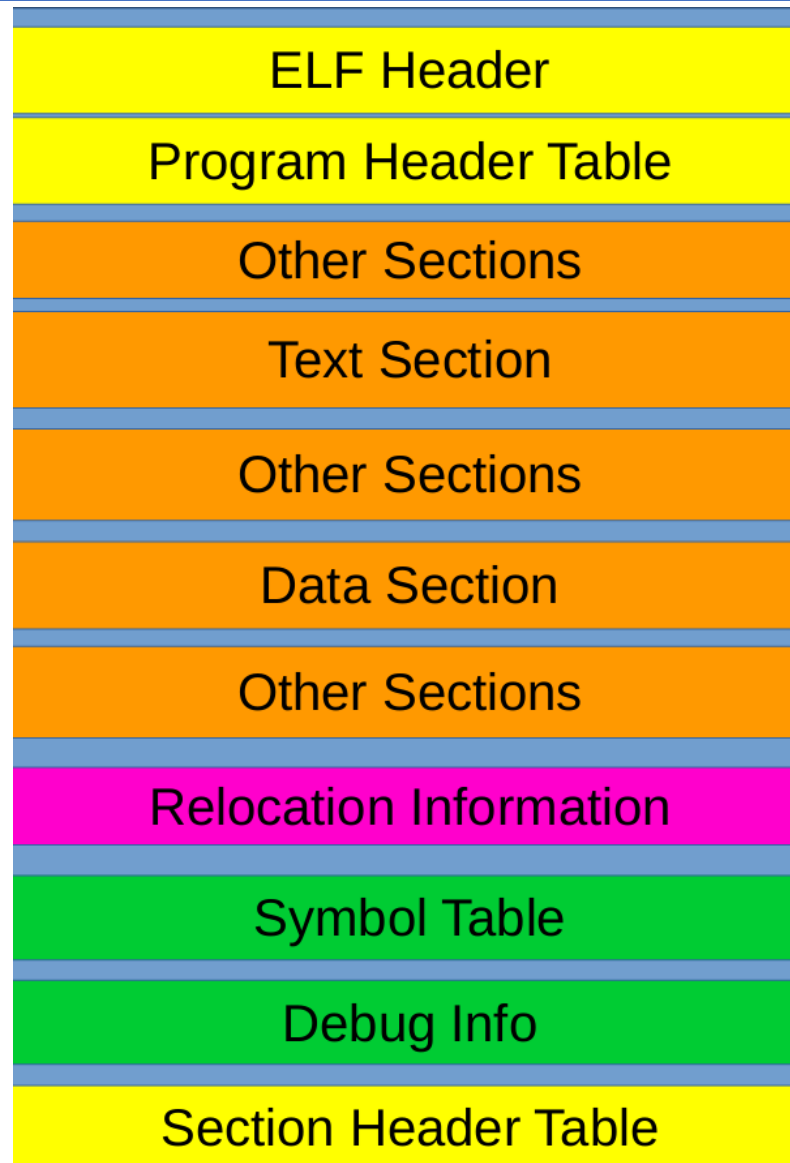
- ELF Header: basic identification information of this file
- Program header table: location of text and data sections
- Text section: the code
- Relocation information: for relocatable text and data sections



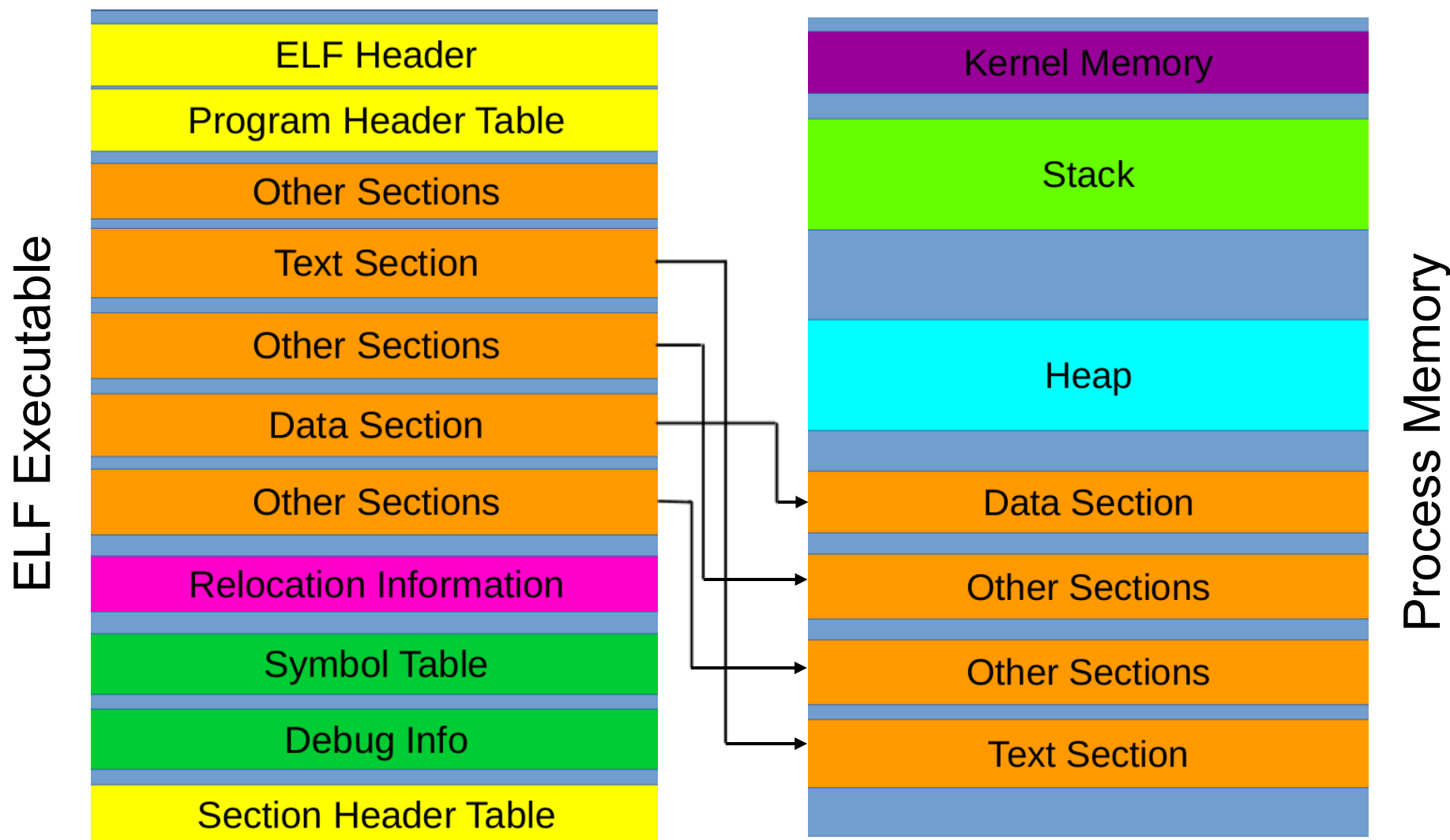
- Data sections:
 - .rodata: read-only
 - .bss: uninitialized global variables
 - .data: initialized global variables
- Example of other sections:
 - .dynamic: dynamic linking information
 - .got: global offset table
 - .init: process initialization code



- Symbol table: locate program symbolic definition (e.g., exported function name)
- Section header table: location and information of each section



ELF File to Process Memory



- Some sections will be directly copied to memory:
 - Example: .text, .data, .init, .dynamic
 - The location (memory addresses) of these sections are defined in the ELF (if not PIC/PIE and ASLR)
- Some sections will not be copied to memory
 - Example: symbol table, debug info

- readelf: Display information about ELF files
 - readelf -h executable
 - Show ELF header

```
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00 00
  Class:                   ELF32
  Data:                     2's complement, little endian
  Version:                  1 (current)
  OS/ABI:                   UNIX - System V
  ABI Version:              0
  Type:                     EXEC (Executable file)
  Machine:                  Intel 80386
  Version:                  0x1
  Entry point address:      0x8048330
```

- readelf -S executable
 - Show section information

			Section address in memory		Section address in file		flag				
[Nr]	Name	Type	Addr		Off	Size	ES	Flg	Lk	Inf	Al
[0]		NULL	00000000		000000	000000	00		0	0	0
[1]	.interp	PROGBITS	08048134		000134	000013	00	A	0	0	1
. . .	.										
[5]	.dynsym	DYNSYM	080481ac		0001ac	000060	10	A	6	1	4
[6]	.dynstr	STRTAB	0804820c		00020c	000053	00	A	0	0	1
. . .	.										
[9]	.rel.dyn	REL	0804828c		00028c	000008	08	A	5	0	4
. . .	.										
[13]	.text	PROGBITS	08048330		000330	0001d2	00	AX	0	0	16
[24]	.data	PROGBITS	08049750		000750	000008	00	WA	0	0	4
[28]	.debug_info	PROGBITS	00000000		00079e	0000d6	00		0	0	1

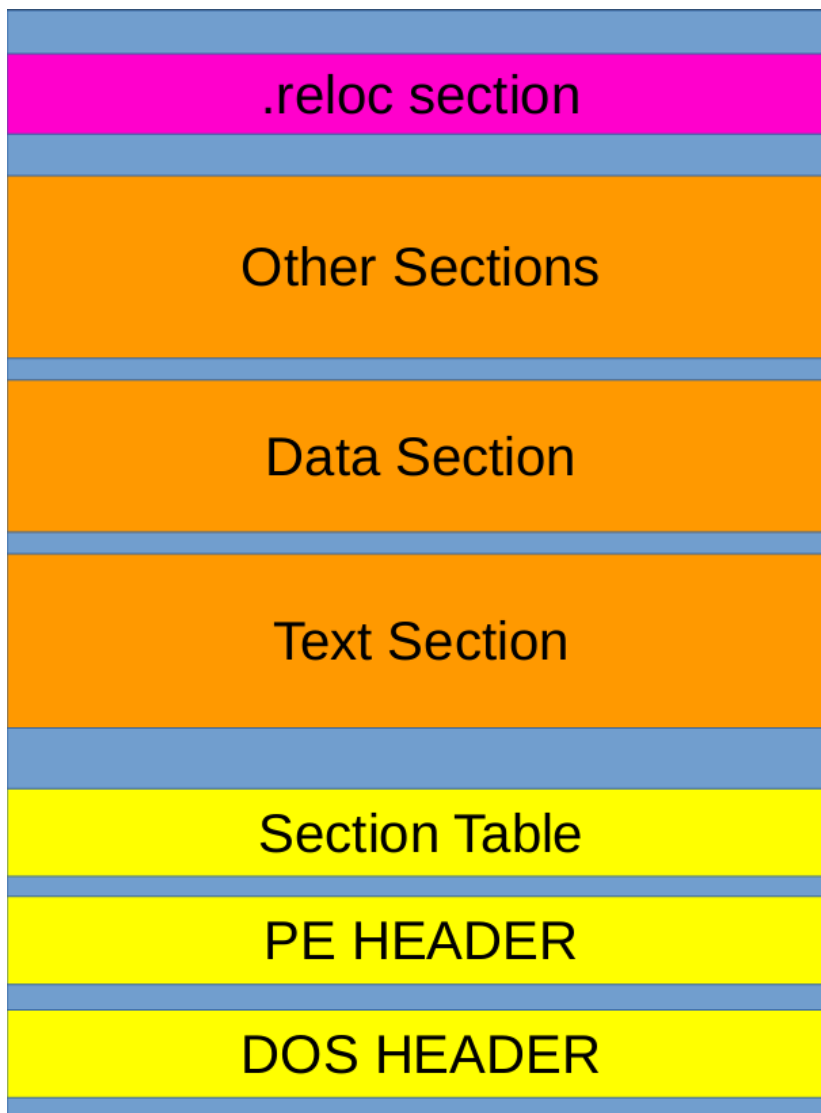
- Each section also has a flag
- In the end of readelf -S output, the flags explained
- The flag bits determine whether a section can be read, written, executed, etc., NOT the section name; viruses might modify the flag bits so that a .text section becomes writable!

- readelf has many other useful options
 - Read the man page for more information
- objdump: the disassembler
- hexdump: raw hexadecimal dump
- file: determine file type
 - file executable
- For more information, Google “ELF format specification”

- Portable Executable
- Also called PE32 (because it is 32-bit code); PE32+ is for 64-bit code
- Older formats exist for 16-bit DOS and Windows 3.1

- Similar to ELF format
 - PE header and DOS header
 - Text and data sections
 - Relocation informations
 - Symbol table
 - Debug information
 - And other sections
- Common sections are .text (for code), .data (read/write data), .rdata (read-only data), .reloc (relocation data used to build IATs)

PE Format Example



- If a program is invoked within a DOS command prompt window, it starts executing here
- For most PE32 executables, the DOS header contains a tiny executable that prints: “This application must be run from Windows”, then exits

- There are empty spaces in executable files
 - The beginning of ELF files
 - Empty spaces between functions
 - Empty spaces between sections
 - Nops in functions
 - Some linkers make executable file align to page boundaries
 - Simplifies the loader's job

- Question: **Why do we care about the details of the PE file format?**
- Answer: Because a virus writer will try to infect the PE file in such a way as to make the virus code execute, while making the PE file look as it would normally look. The job of anti-virus software is to find well disguised viruses.
- Dead spaces are perfect locations to hide viruses
 - CIH virus break itself into parts and hide in the dead spaces between PE sections

- Wei Wang – Defense Against the Dark Arts



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